

EXPERIMENT No. 4

AIM - To understand Continuous Integration, install and configure Jenkins with Maven/Ant/Gradle to setup a build Job

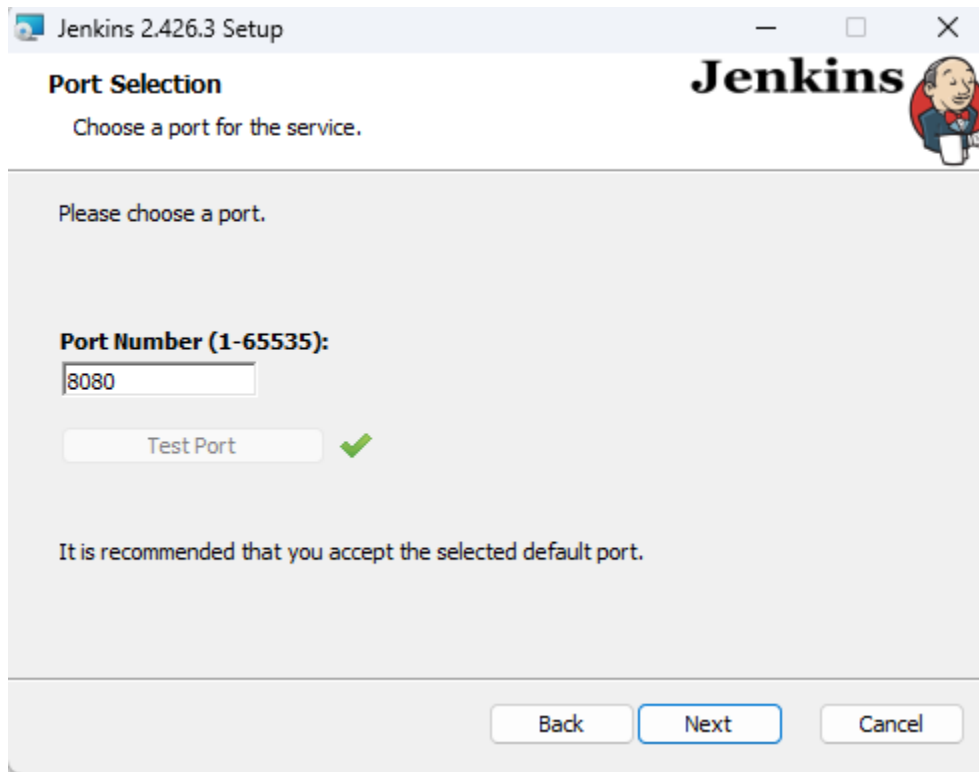
THEORY:

Jenkins is an open-source automation tool written in Java with plugins built for Continuous Integration purpose. Jenkins is used to build and test your software projects continuously making it easier for developers to integrate changes to the project and making it easier for users to obtain a fresh build. It also allows you to continuously deliver your software by integrating with many testing and deployment technologies.

Step 1: Install Jenkins from website



Step 2: Select the desired options



The screenshot shows the 'Jenkins 2.426.3 Setup' window. The title bar includes the window icon, the text 'Jenkins 2.426.3 Setup', and standard minimize, maximize, and close buttons. The window content has a header area with the 'Jenkins' logo and a cartoon character on the right. Below the header, the text 'Port Selection' is followed by the instruction 'Choose a port for the service.' The main area contains the text 'Please choose a port.' and a label 'Port Number (1-65535):' above a text input field containing '8080'. Below the input field is a 'Test Port' button with a green checkmark to its right. At the bottom of the main area, it says 'It is recommended that you accept the selected default port.' The footer contains three buttons: 'Back', 'Next' (which is highlighted with a blue border), and 'Cancel'.

Jenkins 2.426.3 Setup

Port Selection

Choose a port for the service.

Please choose a port.

Port Number (1-65535):

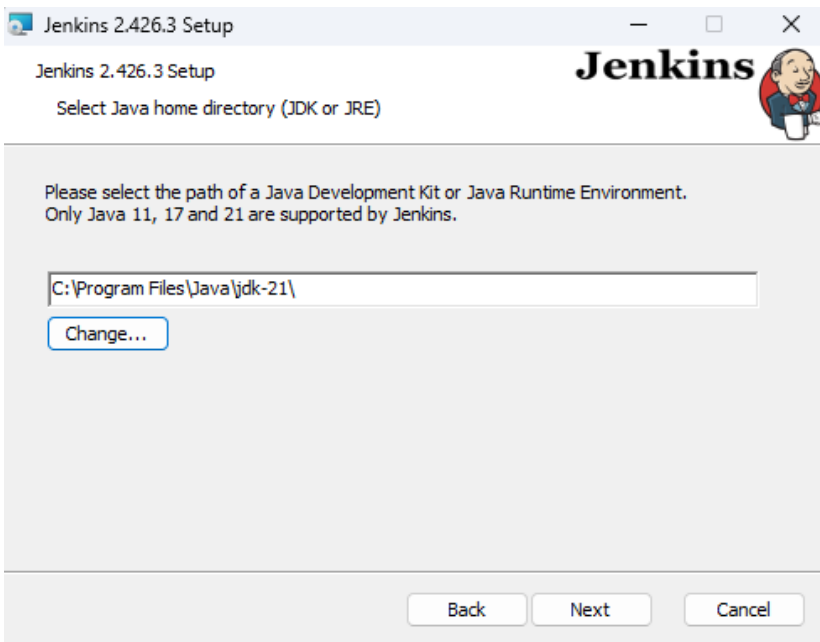
8080

Test Port ✓

It is recommended that you accept the selected default port.

Back Next Cancel

Step 3: Select the files where jdk 21 is installed



The screenshot shows the 'Jenkins 2.426.3 Setup' window. The title bar includes the window icon, the text 'Jenkins 2.426.3 Setup', and standard minimize, maximize, and close buttons. The window content has a header area with the 'Jenkins' logo and a cartoon character on the right. Below the header, the text 'Jenkins 2.426.3 Setup' is followed by the instruction 'Select Java home directory (JDK or JRE)'. The main area contains the text 'Please select the path of a Java Development Kit or Java Runtime Environment. Only Java 11, 17 and 21 are supported by Jenkins.' Below this text is a text input field containing 'C:\Program Files\Java\jdk-21\'. Below the input field is a 'Change...' button. The footer contains three buttons: 'Back', 'Next', and 'Cancel'.

Jenkins 2.426.3 Setup

Jenkins 2.426.3 Setup

Select Java home directory (JDK or JRE)

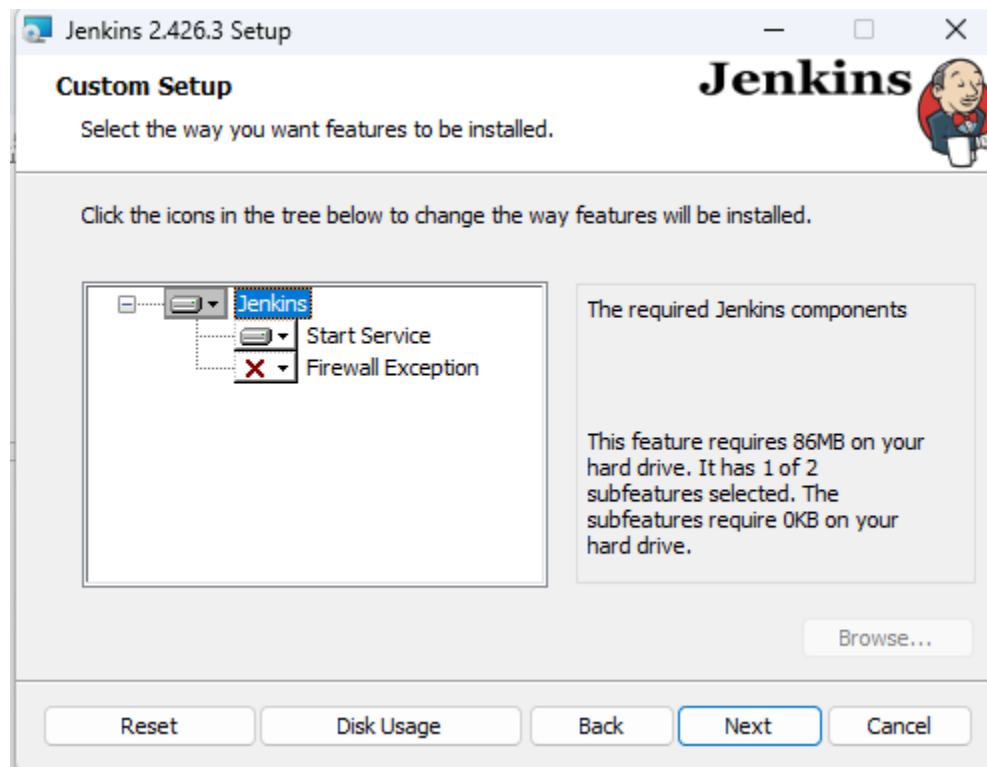
Please select the path of a Java Development Kit or Java Runtime Environment.
Only Java 11, 17 and 21 are supported by Jenkins.

C:\Program Files\Java\jdk-21\

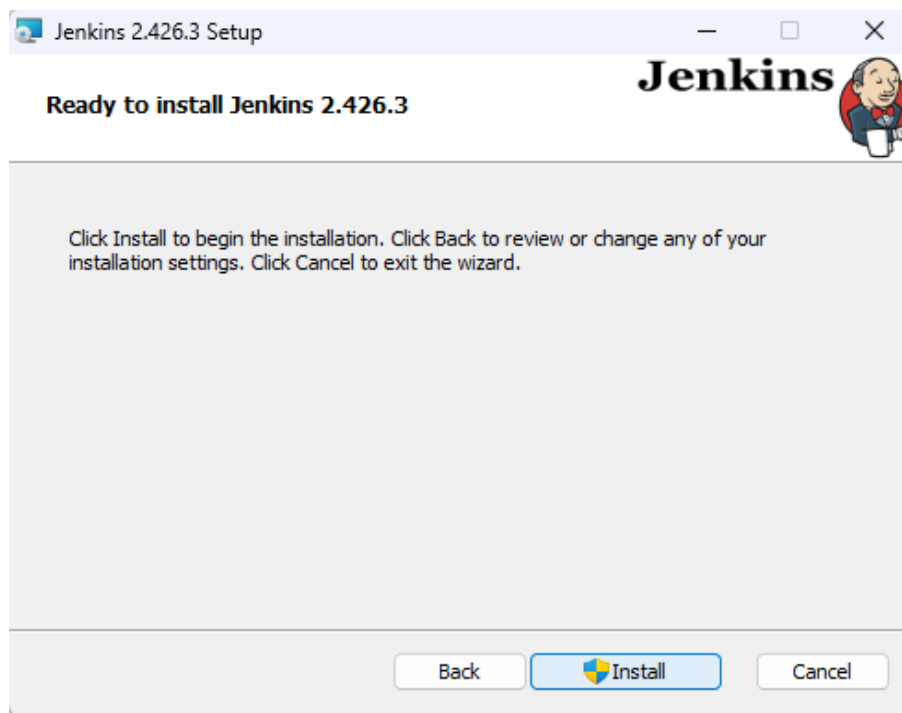
Change...

Back Next Cancel

Step 4: Select the desired options



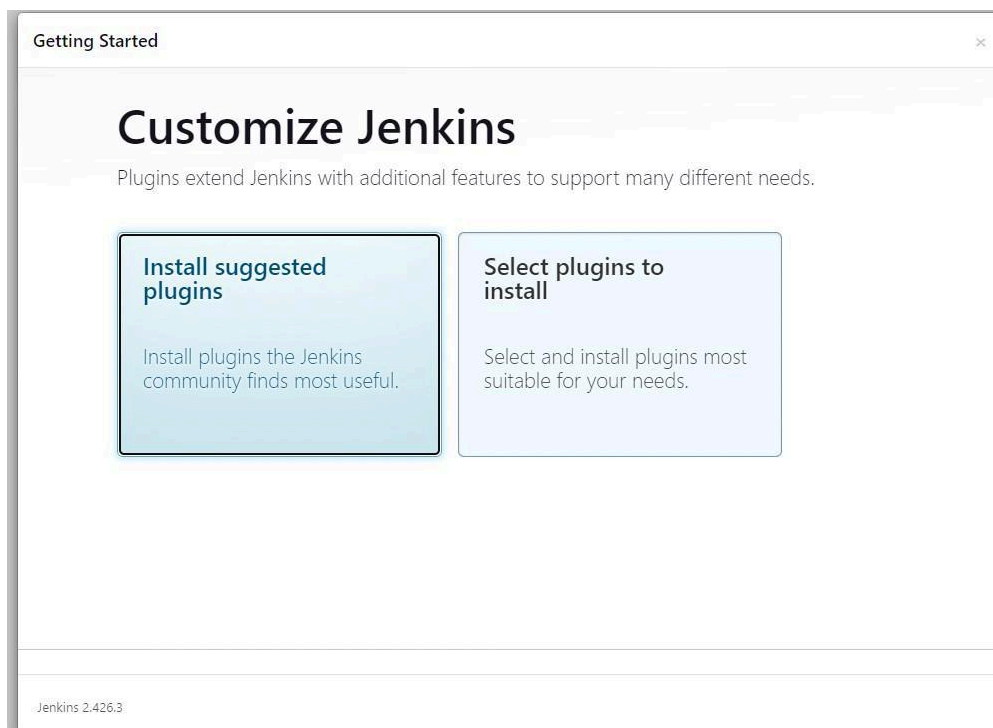
Step 5: Install the file



Step 6: Installation is finished



Step 7: Go on localhost:8080



Step 8: Select the options and wait for a few minutes

Getting Started

Getting Started

☒ Folders	☒ OWASP Markup Formatter	☐ Build Timeout	☐ Credentials Binding	** Icons API
☐ Timestampers	☐ Workspace Cleanup	☐ Ant	☐ Gradle	Folders
☐ Pipeline	☐ GitHub Branch Source	☐ Pipeline: GitHub Groovy Libraries	☐ Pipeline: Stage View	OWASP Markup Formatter
☐ Git	☐ SSH Build Agents	☐ Matrix Authorization Strategy	☐ PAM Authentication	
☐ LDAP	☐ Email Extension	☐ Mailer		

** - required dependency

Jenkins 2.426.3

Step 9: Fill the appropriate credentials

Getting Started

Create First Admin User

Username

Password

Confirm password

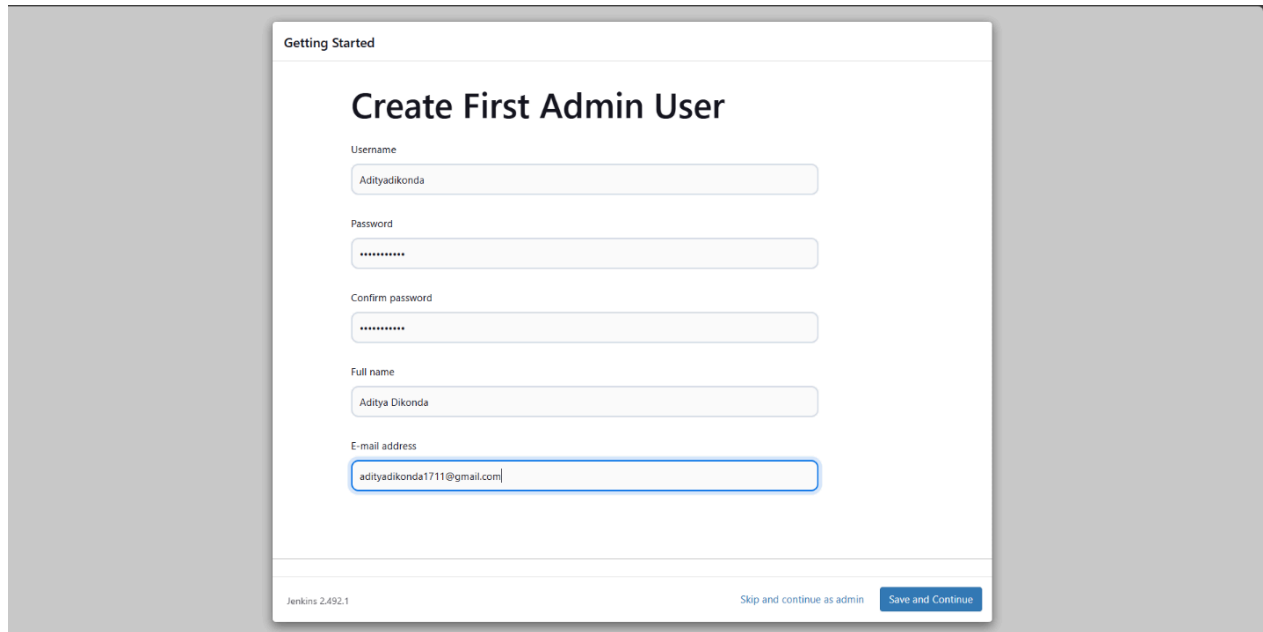
Full name

E-mail address

Jenkins 2.426.3

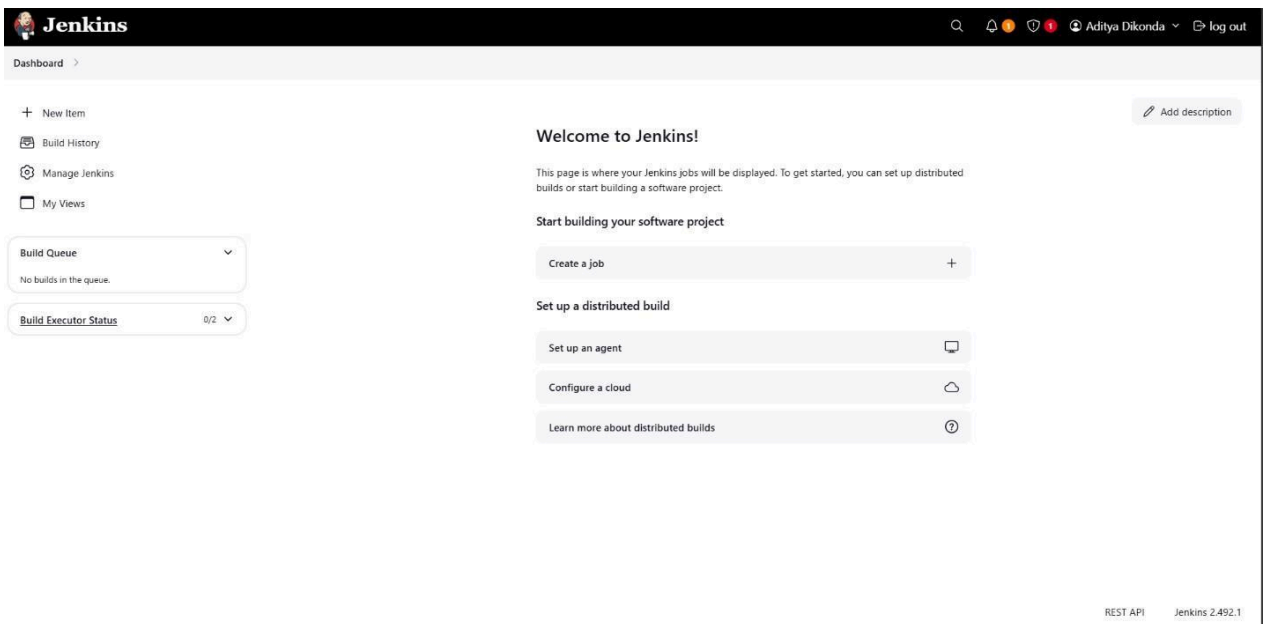
[Skip and continue as admin](#) [Save and Continue](#)

Step 10:



The image shows the Jenkins 'Getting Started' screen. The main heading is 'Create First Admin User'. Below this, there are five input fields: 'Username' (containing 'Adityadikonda'), 'Password' (masked with dots), 'Confirm password' (masked with dots), 'Full name' (containing 'Aditya Dikonda'), and 'E-mail address' (containing 'adityadikonda1711@gmail.com'). At the bottom left, it says 'Jenkins 2.492.1'. At the bottom right, there are two buttons: 'Skip and continue as admin' and 'Save and Continue'.

Step 11: Congratulations! You have created your Jenkins account



The image shows the Jenkins Dashboard. The top navigation bar includes the Jenkins logo, a search icon, a bell icon, a shield icon, a user profile icon for 'Aditya Dikonda', and a 'log out' button. The main content area is divided into two columns. The left column contains a sidebar with links: '+ New Item', 'Build History', 'Manage Jenkins', and 'My Views'. Below these links are two status boxes: 'Build Queue' (showing 'No builds in the queue.') and 'Build Executor Status' (showing '0/2'). The right column features a 'Welcome to Jenkins!' message, followed by a paragraph explaining the dashboard's purpose. Below this, there are three sections: 'Start building your software project' with a 'Create a job' button, 'Set up a distributed build' with buttons for 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. At the bottom right, there is a 'REST API' link and the version 'Jenkins 2.492.1'.

CONCLUSION:

Jenkins with Maven simplifies Continuous Integration, automating builds and improving project efficiency for faster development cycles and higher software quality.